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Physics 225 WFV
Problem Set 1

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Problem 1.1

Revisit our lecture on “relativity without light”. In one of the final steps, it was assumed that

$$\frac{A(v)^2 - 1}{v^2 A(v)^2} = k > 0 \quad (1.1)$$

Explore the cases $k = 0$ and $k < 0$. Are these viable options? Why or why not?

Solution

From the lecture^[1], aside from the derivation of the constant that naturally arises from fundamental assumptions, k .

$$\frac{A^2 - 1}{v^2 A^2} = k \quad (1.2)$$

we also derived the form of the transformation between two frames:¹

$$\begin{bmatrix} t' \\ x' \end{bmatrix} = A \begin{bmatrix} 1 & -kv \\ -v & 1 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix} \quad (1.3)$$

and the value of A as a function of v :

$$A(v) = \frac{1}{\sqrt{1 - kv^2}} \quad (1.4)$$

As well as velocity addition under such transformations

$$w = \frac{u + v}{1 + kuv} \quad (1.5)$$

All of this was under the assumption that $k > 0$. Let’s derive a few relations from this, before actually taking on the prescribed conditions.

For (1.4), we see that the radical $\sqrt{1 - kv^2}$ imposes a restriction on the allowed values of v . The argument of the square root must remain nonnegative:

$$\begin{aligned} 1 - kv^2 &\geq 0, \\ kv^2 &\leq 1, \\ v^2 &\leq \frac{1}{k}, \\ |v| &\leq \frac{1}{\sqrt{k}}. \end{aligned}$$

Thus, for $k > 0$, v cannot exceed $1/\sqrt{k}$ in magnitude. As v approaches this value, the denominator of $A(v)$ approaches zero and therefore $A(v)$ grows without bound.

¹I used (t, x) here instead of the presented coordinates in the lecture, (x, t)

To see what this value implies, we try to add this with the velocity addition formula. Let $u = \frac{1}{\sqrt{k}}$

$$\begin{aligned}
w &= \frac{\frac{1}{\sqrt{k}} + v}{1 + k \frac{1}{\sqrt{k}} v} \\
&= \frac{\frac{1}{\sqrt{k}} + v}{1 + \sqrt{k} v} \\
&= \frac{\frac{1+v\sqrt{k}}{\sqrt{k}}}{1 + \sqrt{k} v} \\
&= \frac{1}{\sqrt{k}} \frac{1 + \sqrt{k} v}{1 + \sqrt{k} v} \\
&= \frac{1}{\sqrt{k}}
\end{aligned} \tag{1.6}$$

Therefore, the value $\frac{1}{\sqrt{k}}$ is invariant. It is a fixed point of the velocity-addition law, and identifies the invariant limiting speed associated with the transformation. Let κ be

$$\kappa = \frac{1}{\sqrt{k}} \tag{1.7}$$

By what we found, κ must be an invariant quantity, and thus, only depends on k , which is itself an arbitrary constant.

Now, we explore what happens when the prescribed conditions are implemented.

(1) $k = 0$

Notice that as we set $k = 0$, (1.4) reduces to

$$A = 1 \tag{1.8}$$

for all v .

Furthermore, the transformation (1.3) becomes

$$\begin{bmatrix} t' \\ x' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -v & 1 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix} \tag{1.9}$$

Additionally, when $k \rightarrow 0^+$ the invariant limiting speed κ tends to infinity.²

It is effectively a shearing operation on the S frame. (See fig. 2)

This is precisely the classical Galilean transformations. Thus, when we set $k = 0$, we recover the Galilean transformation, where the invariant speed is pushed to infinity, and there is no finite invariant speed.

This is a **valid option**, and is in fact the assumption adopted by Newtonian and other classical mechanics theories.

²It cannot be approached from 0^-

(2) $k < 0$

If we set $k < 0$, then (1.4) becomes

$$A(v) = \frac{1}{\sqrt{1 + |k|v^2}} \quad (1.10)$$

The condition that the radicand must be greater than 0 is satisfied for all v , i.e.,

$$\forall v \in \mathbb{R}, k < 0, \quad 1 - kv^2 \geq 0$$

Therefore, v is allowed to grow unbounded. There is also no longer a finite invariant speed. As $|v| \rightarrow \infty$, then $A \rightarrow 0$, which has certain implications on the transformation.

As $v \rightarrow 0$, the limit of (1.3) is still the Galilean transform. However, as $v \rightarrow \infty$, the transformation tends to

$$\begin{bmatrix} t' \\ x' \end{bmatrix} = \begin{bmatrix} 0 & \sqrt{-k} \\ -\frac{1}{\sqrt{-k}} & 0 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix} \quad (1.11)$$

So it's some asymmetric matrix. When it is applied on the event, the proportionality of time and space switches. Time is now proportional to space, and space is proportional to time. While the transformation itself is a valid transformation for finite v , there is no real finite invariant speed.

However,³, the invariant speed becomes

$$\kappa = \frac{1}{\sqrt{k}}$$

But $k < 0$, so the quantity under the square root is negative. Thus, κ is not real and is instead imaginary. Consequently, there is no real invariant speed associated with $k < 0$; the interpretation of κ as a physical invariant speed therefore breaks down.

Thus, $k < 0$ as a Lorentzian transformation invariant is **not a valid option**.

However, as a mathematical transformation, it is perfectly valid.

The transformation itself is well defined.

$$\begin{bmatrix} t' \\ x' \end{bmatrix} = A \begin{bmatrix} 1 & |k|v \\ -v & 1 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

It's a rotation and scaling matrix (See fig. 3). It's invariant is rather this quantity

$$t^2 + |k|x^2$$

And it has a signature of a Euclidean transformation, $(+, +)$. The determinant is 1, and the effect is that it's rotating the axis around the origin such that t', x' is aligned with the t axis and has zero displacement.

Physically, it has no meaning, but geometrically, it is a valid transformation.

³Really, I should have led with this

Below are figures which plot the transformation of the world line from the perspective of S (red), and S' (blue), with the coordinate curves of S (left) superimposed on the reference frame of S' (right). The units are arbitrary. The velocity of S' relative to S is $v = 0.5$, where the invariant speed is normalized, $\kappa \equiv 1$. The yellow lines are the light cones.

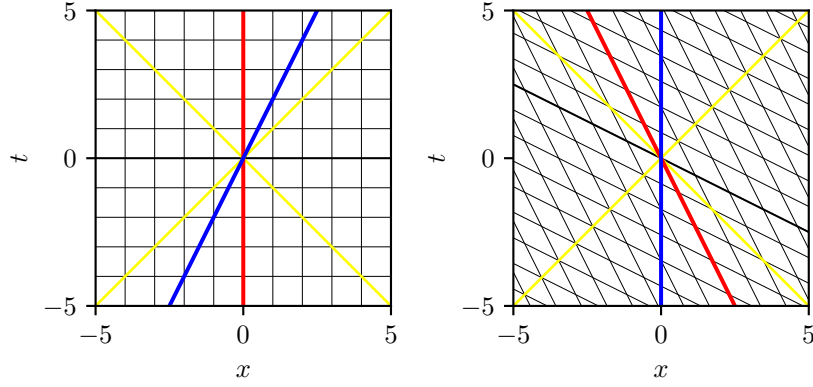


Figure 1: A Lorentz Transformation ($k = 1$)

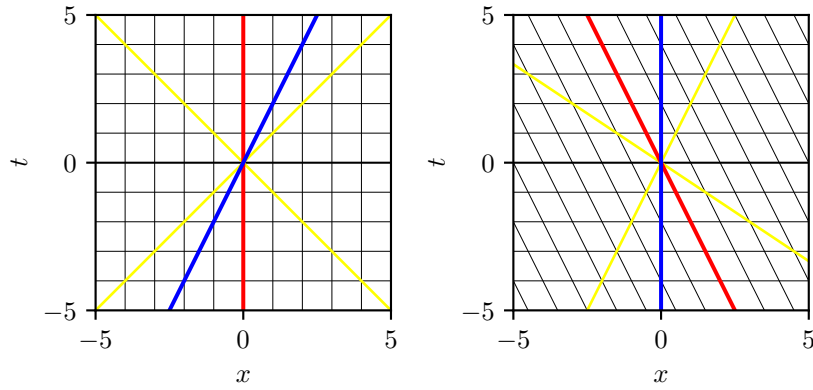


Figure 2: A Galilean Transformation ($k = 0$)

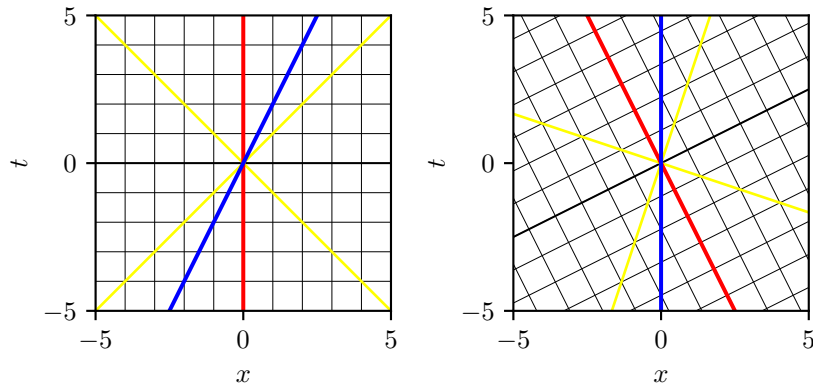


Figure 3: A $k < 0$ Lorentz Transformation ($k = -1$)

Problem 1.2

To get the conditions above, we required closure— i.e., that the composition of two frame transformations $S \rightarrow S'$ and $S' \rightarrow S''$ is also a frame transformation, $S \rightarrow S''$. You will notice that we only used equality of the diagonal components of the composite transformation to get the equation in the problem above. What about the off diagonal components? Explore.

Solution

We again refer to [1]. For a velocity w with respect from reference frame S to S'' , transformation obtained has general form

$$M^\mu_\nu(w) = \begin{bmatrix} A(w) & -vA(w) \\ -\frac{(A(w)^2 - 1)}{vA(v)} & A(w) \end{bmatrix} \quad (2.1)$$

As we imposed from the closure condition, we also require that the composition of a transformation is also a transformation. For velocity v with respect from S to S' , and u , with respect to S' to S'' , we have

$$M^\mu_\sigma(w) = M^\mu_\nu(u)M^\nu_\sigma(v) \quad (2.2)$$

Explicitly showing the multiplication, we have

$$M^\mu_\nu(w) = \begin{bmatrix} A(u) & -uA(u) \\ -\frac{(A(u)^2 - 1)}{uA(u)} & A(u) \end{bmatrix} \begin{bmatrix} A(v) & -vA(v) \\ -\frac{(A(v)^2 - 1)}{vA(v)} & A(v) \end{bmatrix}$$

Solving that, we have the following monstrous matrix multiplication:

$$M^\mu_\nu(w) = \begin{bmatrix} A(u)A(v) + \frac{uA(u)(A(v)^2 - 1)}{vA(v)} & -(u+v)A(u)A(v) \\ -\frac{(A(u)^2 - 1)A(v)}{uA(u)} - \frac{A(u)(A(v)^2 - 1)}{vA(v)} & A(u)A(v) + \frac{vA(v)(A(u)^2 - 1)}{uA(u)} \end{bmatrix}$$

We used the equality of diagonal components, i.e, M^0_0 and M^1_1 , to derive (1.4). Now, we are tasked to determine the implications of the off-diagonal components, i.e., M^0_1 and M^1_0 .

The tricky part here is discerning what it means. The off-diagonal components are not equal, so I can't put a relationship between them, so it took a long time for me to think about what it means. However, I realized that we can again exploit closure. That is, components of $M(w)$ must be equal to components of $M(u)M(v)$. Also, we already know the value of $A(w)$, which is just (1.4)

$$A(w) = \frac{1}{\sqrt{1 - kw^2}}$$

As a preliminary, we begin with the established result on the lecture, that the M_0^1 reduces to

$$\begin{aligned}
-\frac{A(w)^2 - 1}{wA(w)} &= -\frac{\left(\frac{1}{\sqrt{1-kw^2}}\right)^2 - 1}{wA(w)} \\
&= -\frac{\frac{1}{1-kw^2} - 1}{wA(w)} \\
&= -\frac{\frac{1-(1-kw^2)}{1-kw^2}}{wA(w)} \\
&= -\frac{\frac{kw^2}{1-kw^2}}{wA(w)} \\
&= -\frac{kw^2 A(w)^2}{wA(w)} \\
&= -kwA(w)
\end{aligned}$$

However, look at M_0^1 , it is just $-wA(w)$. Hence, we show the relation between these two components:

$$M_1^0 = kM_0^1 \quad (2.3)$$

With that establish, lets begin working on the problem. We start with $M_0^1(w)$. By closure, we require that

$$wA(w) = (u + v)A(u)A(v) \quad (2.4)$$

Likewise, with M_1^0 , we require that

$$\frac{(A(w)^2 - 1)}{wA(w)} = \frac{(A(u)^2 - 1)A(v)}{uA(u)} + \frac{A(u)(A(v)^2 - 1)}{vA(v)} \quad (2.5)$$

We can show that (2.5) just reduces to (2.4)

$$\begin{aligned}
\frac{(A(w)^2 - 1)}{wA(w)} &= \frac{(A(u)^2 - 1)A(v)}{uA(u)} + \frac{A(u)(A(v)^2 - 1)}{vA(v)} \\
\frac{kw^2 A(w)^2}{wA(w)} &= \frac{(ku^2 A(u)^2)A(v)}{uA(u)} + \frac{A(u)(kv^2 A(v)^2)}{vA(v)} \\
kwA(w) &= kvA(u)A(v) + kvA(u)A(v) \\
wA(w) &= (u + v)A(u)A(v)
\end{aligned}$$

What we note here is that (2.4) provides us a consistency condition which determines how A must compose. Now, if we use (1.4), we find that

$$\frac{w}{\sqrt{1-kw^2}} = \frac{u+v}{\sqrt{1-ku^2}\sqrt{1-kv^2}}$$

We massage this a bit and look at what we find. Consider:

$$\begin{aligned}
\frac{w^2}{1 - kw^2} &= \frac{(u + v)^2}{(1 - ku^2)(1 - kv^2)} \\
w^2(1 - ku^2)(1 - kv^2) &= (u + v)^2(1 - kw^2) \\
w^2(1 - ku^2 - kv^2 + k^2u^2v^2) &= (u + v)^2 - kw^2(u + v)^2 \\
w^2(1 - ku^2 - kv^2 + k^2u^2v^2) + kw^2(u + v)^2 &= (u + v)^2 \\
w^2(1 - ku^2 - kv^2 + k^2u^2v^2 + k(u + v)^2) &= (u + v)^2 \\
w^2(1 - ku^2 - kv^2 + k^2u^2v^2 + ku^2 + 2kuv + kv^2) &= (u + v)^2 \\
w^2(1 + 2kuv + k^2u^2v^2) &= (u + v)^2 \\
w^2(1 + kuv)^2 &= (u + v)^2 \\
w^2 &= \frac{(u + v)^2}{(1 + kuv)^2}
\end{aligned}$$

Hence, from the equivalence of $M_0^1(w) = M_0^1(u)M_0^1(v)$, and the from the scaling factor of the components. $M_0^1(w) = kM_1^0(w)$, out comes the relativistic velocity addition law.

$$\boxed{w = \frac{u + v}{1 + kuv}} \quad (2.6)$$

Note that this relation can be independently derived simply from knowing the value of A , and taking the velocities to be

$$w = \frac{dx}{dt}$$

from S to S' , and

$$u = \frac{dx'}{dt'}$$

from S'' to S' . You can then do some math and still arrive with the velocity addition formula.

However, here we see that the notion velocity addition between two moving frames of reference is embedded in the components of the transformation itself.

Problem 1.3

Given a basis for a vector space V , $\{\vec{e}_i\}, i = 1, \dots, n$, we considered the set of n dual vectors $\{\underline{e}^i\}$ according to the rule

$$\underline{e}^i[\vec{e}_j] = \delta^i_j \quad (3.1)$$

- (1) Show that this rule is sufficient to define the $\underline{e}^i$ as dual vectors. That is, we know how each $\underline{e}^i$ acts on all $v \in V$.
- (2) Show that $\{\underline{e}^i\}$ is a basis for V^* .

Solution

Note: I use $\bar{v}$ instead of $\vec{v}$, it's cleaner looking.

- (1) Sufficiency of defining $\underline{e}^i$ as a dual vector.

For the rule $\underline{e}^i[\vec{e}_j] = \delta^i_j$ to define $\underline{e}^i$ as a dual vector, we must know how each $\underline{e}^i$ acts on $\bar{v}$, $\forall \bar{v} \in V$.

By definition, a dual vector is a linear map acting on a vector such that, if $\underline{\omega} \in V^*$, $\bar{v} \in V$, then

$$\underline{\omega} : V \rightarrow \mathbb{R}, \quad \bar{v} \mapsto \underline{\omega}[\bar{v}] \in \mathbb{R}$$

Proof. Let $\bar{v} \in V$. Since $\{\bar{e}_\mu\}$ is a basis of V , there exist unique components $v^\mu \in \mathbb{R}$ such that

$$\bar{v} = v^\mu \bar{e}_\mu.$$

Then

$$\begin{aligned} \underline{e}^\nu[\bar{v}] &= \underline{e}^\nu[v^\mu \bar{e}_\mu] \\ &= v^\mu \underline{e}^\nu[\bar{e}_\mu] \\ &= v^\mu \delta_\mu^\nu \\ &= v^\nu. \end{aligned}$$

Thus, the action of $\underline{e}^\nu$ on every $\bar{v} \in V$ is completely determined. □

- (2) For $\{\underline{e}^i\}$ to be a basis, it must be both linearly independent and spans the entire dual vector space. We prove these two separately.

- (a) Linear Independence.

If $\{\underline{e}^i\}$ is linearly independent, we note that $\forall a_\mu \in \mathbb{R}$, if

$$a_\mu \underline{e}^\mu = \underline{0},$$

then it implies that $\forall \mu, \mu \in (1, \dots, n)$,

$$a_\mu = 0.$$

Proof. Consider a linear combination such that

$$a_\mu \underline{e}^\mu = \underline{0},$$

where $a_\mu \in \mathbb{R}$ are arbitrary.

Evaluating both sides on $e_\nu \in V$ gives

$$\begin{aligned} a_\mu \underline{e}^\mu[\bar{e}_\nu] &= \underline{0}[\bar{e}_\nu] \\ a_\mu \delta_\nu^\mu &= 0 \\ a_\nu &= 0. \end{aligned}$$

Since this holds for every ν , we have $a_\mu = 0$ for all μ . Hence $\{\underline{e}^\mu\}$ is linearly independent. $\square$

(b) Spanning

For $\text{span}\{\underline{e}^i\} = V^*$, it means that $\forall \underline{\omega} \in V^*$, $\underline{\omega}$ can be built as a linear combination of the basis vectors, i.e.,

$$\underline{\omega} = \omega_\mu \underline{e}^\mu$$

Proof. Let $\omega \in V^*$. We define that

$$\omega_\mu = \underline{\omega}[\bar{e}_\mu].$$

For any $\bar{v} \in V$, we can write

$$\bar{v} = v^\mu \bar{e}_\mu.$$

Then

$$\begin{aligned} \underline{\omega}[\bar{v}] &= \underline{\omega}[v^\mu \bar{e}_\mu] \\ &= v^\mu \underline{\omega}[\bar{e}_\mu] \\ &= v^\mu \omega_\mu \\ &= \omega_\mu v^\mu \\ &= \omega_\mu \underline{e}^\mu[v]. \end{aligned}$$

Since v is arbitrary, we can remove it and retain the covector in an operational sense. Therefore,

$$\underline{\omega} = \omega_\mu \underline{e}^\mu,$$

so any $\underline{\omega} \in V^*$ is a linear combination of $\underline{e}^\mu$. Therefore

$$\text{span}\{\underline{e}^\mu\} = V^*.$$

That is, $\underline{e}^\mu$ spans the covector space. $\square$

Hence, $\underline{e}^i$ is a basis for V^* .

Problem 1.4

Show that if $\{\underline{e}^i\}$ is a basis for V^* and $\{\vec{e}_i\}$ is a basis for V , then $\{\vec{e}_i \otimes \underline{e}^j \otimes \underline{e}^k\}$, $i, j, k = 1, \dots, n$, is a basis for the space of $(1, 2)$ tensors.

Solution

To show that $\{\vec{e}_i \otimes \underline{e}^j \otimes \underline{e}^k\}$, where $i, j, k = 1, \dots, n$ is a basis for $(1, 2)$ tensors, we must prove that it is linearly independent and spans the entire $(1, 2)$ tensor space. We shall prove these separately.

Note: I will be using abstract index notation here. Also, I will be switching the indices on the basis vectors and covectors to greek because it makes the math easier.

(1) Linear Independence.

Let 0^a_{bc} be the zero tensor for the $(1, 2)$ tensor space, i.e., it maps all covector and vector arguments to 0.

The set of tensors $\{\vec{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma\}$ is linearly independent if, for all $C_\alpha^{\beta\gamma} \in \mathbb{R}$,

$$C_\alpha^{\beta\gamma} \vec{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma = 0^a_{bc}$$

implies

$$C_\alpha^{\beta\gamma} = 0$$

for all $\alpha, \beta, \gamma = 1, \dots, n$.

Proof. Consider $C_\alpha^{\beta\gamma} \in \mathbb{R}$ such that

$$C_\alpha^{\beta\gamma} \vec{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma = 0^a_{bc}.$$

Let's feed the tensor a set of basis covectors and vectors, $\underline{e}^\lambda, \vec{e}_\mu, \vec{e}_\nu$. Then

$$\begin{aligned} C_\alpha^{\beta\gamma} \vec{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma [\underline{e}^\lambda, \vec{e}_\mu, \vec{e}_\nu] &= 0^a_{bc} [\underline{e}^\lambda, \vec{e}_\mu, \vec{e}_\nu] \\ C_\alpha^{\beta\gamma} \vec{e}_\alpha [\underline{e}^\lambda] \underline{e}^\beta [\vec{e}_\mu] \underline{e}^\gamma [\vec{e}_\nu] &= 0 \\ C_\alpha^{\beta\gamma} \delta^\lambda_\alpha \delta^\beta_\mu \delta^\gamma_\nu &= 0 \\ C_\alpha^{\beta\gamma} \delta^\lambda_\mu \delta^\beta_\nu \delta^\gamma_\alpha &= 0 \\ C_\alpha^{\beta\gamma} \delta^\lambda_\mu \delta^\beta_\alpha \delta^\gamma_\nu &= 0 \\ C_\alpha^{\beta\gamma} \delta^\lambda_\mu \delta^\beta_\nu \delta^\gamma_\alpha &= 0 \\ C_\alpha^{\beta\gamma} \delta^\lambda_\mu \delta^\beta_\alpha \delta^\gamma_\nu &= 0. \end{aligned}$$

Since λ, μ, ν are arbitrary,

$$C_\alpha^{\beta\gamma} = 0$$

for all α, β, γ . □

(2) Spanning

A set of tensors $\{\bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma\}$ span the tensor space $(1, 2)$, if an arbitrary tensor G^a_{bc} can be written as a linear combination of the tensor set $G^\alpha_{\beta\gamma} \in \mathbb{R}, \forall \alpha, \beta, \gamma = 1, \dots, n$, i.e.,

$$G^a_{bc} = G^\alpha_{\beta\gamma} \bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma$$

Proof. We define the following:

For an arbitrary tensor G^a_{bc} , we define that the components are given by⁴

$$G^\alpha_{\beta\gamma} \equiv G^a_{bc}[\underline{e}^\alpha, \bar{e}_\beta, \bar{e}_\gamma]$$

Thus, let's feed the basis covectors and vectors $\underline{e}^\lambda, \bar{e}_\mu, \bar{e}_\nu$

$$\begin{aligned} G^a_{bc}[\underline{e}^\lambda, \bar{e}_\mu, \bar{e}_\nu] &= G^\lambda_{\mu\nu} \\ &= G^\alpha_{\mu\nu} \delta^\lambda_\alpha \\ &= G^\alpha_{\beta\nu} \delta^\lambda_\alpha \delta^\beta_\mu \\ &= G^\alpha_{\beta\nu} \delta^\lambda_\alpha \delta^\beta_\mu \delta^\gamma_\nu \\ &= G^\alpha_{\beta\nu} \bar{e}_\alpha[\underline{e}^\lambda] \underline{e}^\beta[\bar{e}_\mu] \underline{e}^\gamma[\bar{e}_\nu] \\ &= G^\lambda_{\mu\nu} (\bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma)[\underline{e}^\lambda, \bar{e}_\mu, \bar{e}_\nu] \end{aligned}$$

However, the term in the parenthesis is just the tensor set, thus

$$G^a_{bc}[\underline{e}^\alpha, \bar{e}_\beta, \bar{e}_\gamma] = G^\lambda_{\mu\nu} \bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma[\underline{e}^\lambda, \bar{e}_\mu, \bar{e}_\nu]$$

Since the arguments $\underline{e}^\lambda, \bar{e}_\mu, \bar{e}_\nu$ are arbitrary, we can remove them and retain the tensor as an operator. Therefore

$$\begin{aligned} G^a_{bc} &= G^\alpha_{\beta\gamma} \bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma \\ \implies \text{span}\{\bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma\} &= \binom{1}{2} \end{aligned}$$

Thus, the tensors

$$\{\bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma\}$$

span the $(1, 2)$ tensor space. □

Hence, $\{\bar{e}_\alpha \otimes \underline{e}^\beta \otimes \underline{e}^\gamma\}$ constitute a basis for the $(1, 2)$ tensor space.

⁴I kinda feel like this move is a hack of some sort. Feels like a circular argument without really having justification. I don't understand why we define it to be like that (other than to define it in terms of the basis itself), but the lecture defined things like this multiple times so yeah, just kinda followed it.

Problem 1.5

If g is a $(0, 2)$ tensor, then it defines an isomorphism between V and V^* , i.e., $g : V \rightarrow V^*$. Show that the inverse map $g^{-1} : V^* \rightarrow V$, such that $g^{-1}g = gg^{-1} = \text{Id}$, is a $(2, 0)$ tensor.

Solution

Proof. A $(2, 0)$ tensor is a bilinear map such that, if $\alpha \in (2, 0)$, $\omega, \eta \in V^*$

$$\alpha : V^* \times V^* \rightarrow \mathbb{R}, \quad \alpha : \omega, \eta \mapsto \alpha(\omega, \eta).$$

Now we figure out how to massage g^{-1} to show this property. Since $g \in (0, 2)$ it accepts two vectors and returns a scalar. However, it can also only accept one, and can be defined as a map such that

$$g : V \rightarrow V^*.$$

We have the inverse map be

$$g^{-1} : V^* \rightarrow V.$$

That is, it maps covectors to vectors. Let $\omega, \eta \in V^*$, then

$$g^{-1}(\omega) \in V.$$

Since η is a covector, it's a map from $V \rightarrow \mathbb{R}$. Hence, we can have η act on $g^{-1}(\omega)$,

$$\eta(g^{-1}(\omega)) \in \mathbb{R}.$$

Let us define g^{-1} to accept two arguments, such that

$$g^{-1}(\omega, \eta) \equiv \eta(g^{-1}(\omega)) \in \mathbb{R}.$$

Thus, we have shown that g^{-1} can be a map such that

$$g^{-1} : V^* \times V^* \rightarrow \mathbb{R}.$$

However, this is not enough to include it in the group of $(2, 0)$ tensors.

Since $g^{-1} : V^* \rightarrow V$ is an isomorphism, it implies linearity between covectors, i.e.,

$$g^{-1}(a\omega_1 + b\omega_2) = ag^{-1}(\omega_1) + bg^{-1}(\omega_2).$$

Let us check bilinearity for two covectors.

For $a, b \in \mathbb{R}$, then

$$\begin{aligned} g^{-1}(a\omega_1 + b\omega_2, \eta) &= \eta(g^{-1}(a\omega_1 + b\omega_2)) \\ &= \eta(g^{-1}(a\omega_1) + g^{-1}(b\omega_2)) \\ &= a\eta g^{-1}(\omega_1) + b\eta g^{-1}(\omega_2) \\ &= ag^{-1}(\omega_1, \eta) + bg^{-1}(\omega_2, \eta). \end{aligned}$$

Similarly,

$$g^{-1}(\omega, a\eta_1 + b\eta_2) = ag^{-1}(\omega, \eta_1) + bg^{-1}(\omega, \eta_2).$$

Thus, g^{-1} is a bilinear on $V^* \times V^*$, and therefore, a $(2, 0)$ tensor. $\square$

No longer part of the main proof, but a less rigorous but quicker argument is this: Since g is defined to be

$$g = g_{\mu\nu} \partial_\mu \otimes \partial_\nu.$$

Then its inverse, such that

$$gg^{-1} = g^{-1}g = \delta,$$

takes the form

$$g^{-1} = g^{\mu\nu} dx^\mu \otimes dx^\nu,$$

where

$$g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu_\nu.$$

Hence, g^{-1} is a $(2, 0)$ tensor with components $g^{\mu\nu}$.

Problem 1.6

Starting from the definition of a tensor as a multilinear map, prove the standard transformation law for (k, l) -tensor components under a change of coordinates.

Solution

The standard Tensor component transformation law under change of coordinates from $\{x^\sigma\}$ to $\{x'^\mu\}$ is [1, 2]

$$T'^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} = \frac{\partial x'^{\mu_1}}{\partial x^{\rho_1}} \dots \frac{\partial x'^{\mu_k}}{\partial x^{\rho_k}} \frac{\partial x^{\sigma_1}}{\partial x'^{\nu_1}} \dots \frac{\partial x^{\sigma_l}}{\partial x'^{\nu_l}} T^{\rho_1 \dots \rho_k}_{\sigma_1 \dots \sigma_l} \quad (6.1)$$

We derive it, from the definition alone, as follows:

Proof. The definition of a tensor, as stated from the lecture, is [1]:

A tensor of (k, l) type is a multilinear map from the product of dual and vector spaces to real numbers, such that

$$T : V_1^* \times V_2^* \times \dots \times V_k^* \times V_1 \times V_2 \times \dots \times V_l \rightarrow \mathbb{R} \quad (6.2)$$

That is, it is a *machine* which accepts covectors (k slots), and vectors (l slots), and returns a real number. Multilinearity here means that the tensor is linear in each slot separately.

Vectors are tensors which accept one covector, and likewise covectors are tensors which accept one vector. Hence, we can use basis vectors and basis covectors as a basis for a tensor space. Let $\{x^\mu\}$ be coordinate functions in a manifold $\mathcal{M}$, and $T \in (k, l)$, $\omega_i \in V^*$, $v^i \in V$ such that we can define vectors and covectors naturally:

$$v = v^\mu \partial_\mu, \quad \omega = \omega_\mu dx^\mu$$

Therefore, the basis vectors are ∂_μ , and the basis covectors are dx^μ .

We define that if we feed the tensor basis vectors and covectors, we recover its components.

$$T[dx^{\mu_1}, \dots, dx^{\mu_k}; \partial_{\nu_1}, \dots, \partial_{\nu_l}] \equiv T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l}$$

We can then expand our way into $k + l$ identity $\delta_{\nu_l}^{\mu_k}$ operations, pulling out the basis for each index, separating them, discovering it's an outer product, and recovering the tensor as an operational definition (see the solution on Problem 1.4.2 which uses this method).

With all that done, we find that a tensor $T \in (k, l)$ defined in a chart $\{x^\mu\}$ has component form, which is the form that we need to define the transformation in general.

$$T = T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} \partial_{\mu_1} \otimes \dots \otimes \partial_{\mu_k} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_l} \quad (6.3)$$

Say we wish to change our coordinates from $\{x^\mu\}$ to $\{y^\sigma\}$. Recall how a vector components transform under a change of coordinates:

$$\frac{\partial}{\partial y^\sigma} = \frac{\partial x^\mu}{\partial y^\sigma} \frac{\partial}{\partial x^\mu} \quad (6.4)$$

Additionally, recall how a covector components transform under a change of coordinates

$$dy^\sigma = \frac{\partial y^\sigma}{\partial x^\mu} dx^\mu \quad (6.5)$$

Let's rewrite the expression in terms of this change of coordinates. We will now explicitly write $\partial_\mu = \frac{\partial}{\partial x^\mu}$, since there are multiple coordinates involved.

We're going from x^μ to y^σ , so it's gonna be the inverse of the expressions above.

$$T = T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} \left(\frac{\partial y^{\rho_1}}{\partial x^{\mu_1}} \frac{\partial}{\partial y^{\rho_1}} \right) \otimes \dots \otimes \left(\frac{\partial y^{\rho_k}}{\partial x^{\mu_k}} \frac{\partial}{\partial y^{\rho_k}} \right) \otimes \left(\frac{\partial x^{\nu_1}}{\partial y^{\sigma_1}} dy^{\sigma_1} \right) \otimes \left(\frac{\partial x^{\nu_l}}{\partial y^{\sigma_l}} dy^{\sigma_l} \right)$$

We can pull out the coordinate transformation coefficients, a.k.a. the Jacobians, they're just constants, and the tensor product is a linear operation.

$$T = \frac{\partial y^{\rho_1}}{\partial x^{\mu_1}} \dots \frac{\partial y^{\rho_k}}{\partial x^{\mu_k}} \frac{\partial x^{\nu_1}}{\partial y^{\sigma_1}} \dots \frac{\partial x^{\nu_l}}{\partial y^{\sigma_l}} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l} \frac{\partial}{\partial y^{\rho_1}} \otimes \dots \otimes \frac{\partial}{\partial y^{\rho_k}} \otimes dy^{\sigma_1} \otimes \dots \otimes dy^{\sigma_l} \quad (6.6)$$

On the other hand, since T is the same tensor regardless of the coordinate system, we can write it in the terms of the $\{y^\sigma\}$ coordinates directly. We denote the components for this new coordinates to be T'

$$T = T'^{\rho_1 \dots \rho_k}_{\sigma_1 \dots \sigma_l} \frac{\partial}{\partial y^{\rho_1}} \otimes \dots \otimes \frac{\partial}{\partial y^{\rho_k}} \otimes dy^{\sigma_1} \otimes \dots \otimes dy^{\sigma_l} \quad (6.7)$$

Let's compare (6.6) and (6.7). They are equal, they have the same basis tensors. However, their components are not the same. But since both statements are equal, then the components of both forms must also be equal, only transformations from one another. Hence, we equate them, and find the following result:

$$\boxed{T'^{\rho_1 \dots \rho_k}_{\sigma_1 \dots \sigma_l} = \frac{\partial y^{\rho_1}}{\partial x^{\mu_1}} \dots \frac{\partial y^{\rho_k}}{\partial x^{\mu_k}} \frac{\partial x^{\nu_1}}{\partial y^{\sigma_1}} \dots \frac{\partial x^{\nu_l}}{\partial y^{\sigma_l}} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l}} \quad (6.8)$$

Therefore, we have recovered (6.1),⁵ which is how a component of a (k, l) tensor transforms under a coordinate transformation. $\square$

⁵With some caveats, it used x' instead of y , which isn't really a problem since we can just relabel it, and the indices are swapped, but indices are arbitrary, and the swapping is consistent. We can just rename it and swap them, but I am too tired, and I neither have the drive nor inclination to do so. It is equivalent to Eq. (6.1). Trust me bro.

Problem 1.7

Show that the identity maps

$$\text{Id}_1 : V \rightarrow V, \text{Id}_1(v) = v, \forall v \in V \quad (7.1)$$

$$\text{Id}_2 : V^* \rightarrow V^*, \text{Id}_2(\omega) = \omega, \forall \omega \in V^* \quad (7.2)$$

Can be interpreted as the same $(1, 1)$ tensor. What are its components? What does this tensor do when it is fed a one-form and a vector? Does this remind you of any other tensorial operation?

Solution

We tackle this one first with a functional approach, then we do the quicker coordinate-based route.

Proof. A $(1, 1)$ identity tensor can be viewed as a map

$$V \rightarrow V, \quad V^* \rightarrow V^*$$

or equivalently, as a bilinear map

$$V^* \times V \rightarrow \mathbb{R}$$

We start with $\text{Id}_1 : V \rightarrow V$.

For any $v \in V$, then

$$\text{Id}_1[v] = v.$$

We can turn this into a scalar-valued operation by feeding the resulting vector to a one-form $\omega \in V^*$:

$$\omega[\text{Id}_1[v]] = \omega[v]$$

As we did in Problem 1.5, we can define this as a bilinear map, a $(1, 1)$ tensor call it δ^6 , such that

$$\delta(\omega, v) \equiv \omega[v]$$

For $\text{Id}_2 : V^* \rightarrow V^*$, given $\omega \in V^*$, then

$$\text{Id}_2(\omega) = \omega$$

A vector v naturally acts on a one-form by evaluation. Alternatively, we could use the fact that the double-dual space V^{**} is isomorphic to V , $V^{**} \cong V$, so

$$\text{Id}_2(\omega)[v] \equiv v[\text{Id}_2(\omega)] = \omega(v)$$

Again, we find that

$$\delta(\omega, v) = \omega[v]$$

Both identity maps map v and ω to the same scalar value. Hence, it is the same $(1, 1)$ tensor. □

⁶wow, what a shocker choice of name

Alternatively, since it a vector and a covector can be represented with a vector and covector basis,

$$v = v^\mu \bar{e}_\mu, \quad \omega = \omega_\nu \underline{e}^\nu$$

Then we have

$$\begin{aligned} \delta(\omega, v) &= \delta[v^\mu \bar{e}_\mu, \omega_\nu \underline{e}^\nu] \\ &= v^\mu \omega_\nu \delta[\bar{e}_\mu, \underline{e}^\nu] \\ &= v^\mu \omega_\nu \delta^\mu_\nu \\ &= v^\mu \omega_\mu \end{aligned}$$

That is, it is just the Euclidean tensor/metric, or the Kronecker delta, or whatever other name it has.

Thus, its components are

$$\delta^\mu_\nu = \begin{cases} 1, & \mu = \nu \\ 0, & \mu \neq \nu \end{cases}$$

Since it is a $(1, 1)$ tensor, and matrices are isomorphic to $(1, 1)$ tensors, it can be represented as a matrix

$$\delta^\mu_\nu = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

When fed a one-form and a vector, it simply evaluates the one-form on the vector

$$\delta(\omega, v) = \omega(v)$$

Otherwise known as a contraction operation, where we contract the indices upstairs and downstairs.

$$\delta^\mu_\nu \omega_\mu v^\nu = \omega_\nu v^\nu = \omega(v)$$

So the identity $(1, 1)$ tensor is just the Euclidean metric, or the Kronecker delta, or the identity under contraction operator. Among other names.

References

- [1] Vercil Juan. Lecture Notes on General Relativity: Physics 225, August 2026. *Based on lectures delivered by Dr. Ian Vega, 1st Semester, AY 2026–2027.* URL: https://github.com/silvernuke911/Class_Notes/blob/main/Physics%20225/p225.pdf.
- [2] Canbin Liang and Bin Zhou. *Differential Geometry and General Relativity*. Graduate Texts in Physics. Springer, Singapore, 2023. doi:10.1007/978-981-99-0022-0.

Appendices

Code

No code was used for this problem set

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